



Project Report On

Dark Web Threat Monitoring Crawler

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CERTIFICATE

*This is to certify that the project report entitled “**Dark Web Threat Monitoring Crawler**” is a bonafide record of the work done by **Diya Vypana (U2203081)**, **Jithin Shabu (U2203114)**, **Levana P Saju (U2203133)** and **Mathew Dominic (U2203138)**, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in **Computer Science and Engineering** during the academic year **2025–2026**.*

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Abstract

The increasing presence of stolen credentials, phishing kits, and impersonation activities on the dark web has created a significant challenge for organizations, as conventional cybersecurity tools lack visibility into hidden Tor networks. This project presents a *Dark Web Threat Monitoring Crawler*, a system designed to automatically discover and analyze potential threats related to organizational data on .onion websites.

The system employs a Tor-enabled web crawler to collect publicly accessible dark web pages using keyword-based searches. The collected data is processed through a preprocessing pipeline that removes noise, extracts relevant text, and prepares it for analysis. Threat detection is performed using a hybrid approach that combines rule-based pattern matching with a transformer-based machine learning model (DarkBERT) to identify indicators such as credential leaks, phishing content, and financial information exposure.

Detected threats are classified according to severity levels and stored in a structured database for further analysis. The results are visualized through an interactive Streamlit dashboard that provides threat summaries, evidence snippets, and exportable reports for security monitoring.

Compared to traditional monitoring systems, the proposed approach enables automated discovery of hidden threats on the dark web and provides organizations with actionable intelligence to detect potential risks early and strengthen their cybersecurity preparedness.

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Chapter 1

Introduction

This chapter introduces the Dark Web Threat Monitoring Crawler, a system designed to detect potential cybersecurity threats originating from hidden web networks. It explains the limitations of traditional cybersecurity monitoring tools that primarily focus on surface web sources and internal network activity, highlighting the need for automated monitoring mechanisms capable of analyzing dark web content. The chapter presents the background of dark web environments and modern cyber threat intelligence approaches, defines the main problem addressed in this project, and establishes the foundation for the proposed system. Subsequent sections describe the objectives, scope, challenges, assumptions, and relevance of the developed framework.

1.1 Background

The rapid expansion of the internet has significantly increased the amount of digital information available online. Alongside the surface web, which is indexed by conventional search engines, there exists a hidden layer known as the dark web. The dark web operates through anonymity networks such as Tor, which allow users to access services and communicate without revealing their identities or locations. While these networks were originally designed to support privacy and freedom of expression, they have also become platforms for various illicit activities including the distribution of stolen credentials, phishing infrastructure, leaked databases, and illegal marketplaces.

Modern cybersecurity systems primarily focus on monitoring surface web activity, enterprise networks, and internal security logs. However, threats that originate from the dark web often remain undetected until significant damage has already occurred. Cybercriminal communities frequently use hidden forums and marketplaces to exchange compromised data, sell access to breached systems, or coordinate cyberattacks. As a

result, organizations require effective tools that can monitor dark web platforms and identify early indicators of potential security threats.

Automated crawling and analysis systems have therefore become increasingly important for cyber threat intelligence. By collecting and analyzing publicly accessible dark web content, such systems can help organizations detect leaked credentials, phishing campaigns, or brand impersonation attempts at an early stage. Integrating web crawling, natural language processing, and machine learning techniques enables the development of intelligent monitoring systems capable of extracting meaningful insights from unstructured dark web data.

1.2 Problem Definition

The central problem addressed in this project is the lack of effective automated systems capable of detecting cyber threats originating from dark web platforms. Traditional cybersecurity monitoring tools are designed primarily for surface web environments and internal network infrastructure, making them ineffective in identifying malicious activities occurring within anonymous networks such as Tor.

Dark web platforms frequently host information related to stolen credentials, phishing kits, and compromised databases that may directly impact organizations. However, monitoring these environments manually is extremely difficult due to the constantly changing nature of .onion websites, the large volume of unstructured data, and the anonymity mechanisms used within these networks.

Therefore, the problem is to design an automated framework capable of securely crawling dark web pages, extracting meaningful textual information, identifying potential cybersecurity threats, and presenting actionable insights to security analysts. Addressing this problem requires combining Tor-enabled web crawling, natural language processing techniques, and machine learning-based threat classification into a unified monitoring system.

1.3 Scope and Motivation

Scope

The scope of this project is to design and implement a system capable of automatically crawling publicly accessible dark web pages and analyzing their content to detect potential cybersecurity threats related to organizations. The system integrates Tor-enabled focused crawling, text preprocessing techniques, rule-based threat detection mechanisms, and transformer-based machine learning models for threat classification.

The system focuses on analyzing textual information extracted from dark web pages and identifying indicators of threats such as credential leaks, phishing infrastructure, and impersonation attempts. Detected threats are stored in a structured database and visualized through an interactive dashboard that allows users to monitor emerging threats and generate reports. The scope of the project is limited to threat detection and monitoring and does not involve direct intervention or disruption of malicious activities on the dark web.

Motivation

The motivation for this project arises from the increasing number of cybersecurity threats emerging from hidden online environments. Sensitive information such as login credentials, personal data, and financial records is frequently leaked or traded on dark web platforms. Organizations often become aware of such breaches only after significant damage has occurred.

Traditional security monitoring approaches lack visibility into these hidden networks and therefore fail to detect early indicators of cyberattacks. Automated dark web monitoring systems can provide valuable threat intelligence by continuously scanning hidden platforms and identifying suspicious content related to organizations.

By integrating web crawling, natural language processing, and machine learning techniques, this project aims to develop an intelligent monitoring framework capable of identifying potential threats at an early stage. Such a system can assist cybersecurity teams in improving situational awareness and responding proactively to emerging cyber threats.

1.4 Objectives

In order to address the challenges of monitoring cyber threats originating from dark web environments, the following objectives are defined:

- **Develop a Tor-Enabled Dark Web Crawler** Design a focused crawler capable of securely accessing and collecting data from .onion websites through the Tor network.
- **Implement a Preprocessing Pipeline for Dark Web Content** Extract and normalize textual information from raw HTML pages by removing scripts, tags, and irrelevant content.
- **Detect Cyber Threat Indicators** Identify patterns associated with credential leaks, phishing portals, and compromised databases using rule-based detection techniques.
- **Apply Machine Learning for Threat Classification** Utilize transformer-based models such as DarkBERT to classify crawled content into categories such as credential leak, phishing, impersonation, or benign content.
- **Design a Threat Severity Assessment Mechanism** Develop a severity understanding model that prioritizes threats based on contextual signals such as keyword presence, model confidence, and recency.
- **Provide an Interactive Visualization Dashboard** Present detected threats and analytics through an interactive dashboard that supports monitoring, filtering, and report generation.

1.5 Challenges

Developing a dark web monitoring system presents several technical challenges. First, dark web websites frequently change URLs or disappear, making consistent crawling difficult. Second, collected data often contains noisy or unstructured content that must be carefully processed before meaningful analysis can be performed.

Another challenge lies in accurately identifying malicious content from large volumes of textual data. Rule-based detection methods may produce false positives, while machine

learning models require careful design to maintain accuracy without excessive computational cost. Additionally, maintaining ethical and legal boundaries while crawling dark web content requires careful system design and controlled data collection strategies.

1.6 Assumptions

The proposed system is designed under several assumptions regarding the operational environment and system design. First, it is assumed that the Tor network remains accessible and stable for retrieving publicly available .onion web pages. Second, the system assumes that relevant dark web content can be discovered using keyword-based search methods such as Ahmia indexing.

It is also assumed that textual information extracted from dark web pages contains sufficient contextual indicators to identify potential cybersecurity threats. Finally, the system assumes that crawling activities remain within ethical and legal boundaries by focusing only on publicly accessible content and avoiding unauthorized interactions with restricted systems.

1.7 Societal / Industrial Relevance

The Dark Web Threat Monitoring Crawler has significant relevance for both cybersecurity operations and digital risk management. Organizations can use such systems to detect leaked credentials, phishing campaigns, or brand impersonation attempts before they escalate into large-scale security incidents.

From an industrial perspective, automated dark web monitoring enhances cyber threat intelligence capabilities and supports proactive defense strategies. Security analysts can leverage insights generated by the system to strengthen incident response and improve organizational resilience against cyberattacks.

From a societal perspective, early detection of cybercrime activities contributes to a safer digital ecosystem by reducing the spread of stolen data and malicious services. Systems capable of monitoring hidden web environments therefore play an important role in strengthening cybersecurity awareness and protecting sensitive information.

1.8 Organization of the Report

This report is organized into the following chapters:

- **Chapter 1: Introduction** – Introduces the background of dark web environments, defines the problem of detecting cybersecurity threats from hidden networks, and outlines the objectives, scope, challenges, assumptions, and relevance of the proposed system.
- **Chapter 2: Literature Review** – Presents a survey of existing research related to dark web crawling, cyber threat intelligence systems, and machine learning techniques used for threat detection.
- **Chapter 3: Proposed Methodology** – Describes the overall methodology and architecture of the Dark Web Threat Monitoring Crawler, including the data acquisition layer, preprocessing pipeline, threat detection mechanisms, and visualization components.
- **Chapter 4: System Design and Implementation** – Explains the design of the system modules and discusses the implementation of the Tor-enabled crawler, preprocessing pipeline, threat detection system, and visualization dashboard.
- **Chapter 5: Results and Discussions** – Presents the experimental results obtained from the implemented system and discusses the performance and effectiveness of the proposed dark web monitoring framework.
- **Chapter 6: Conclusions and Future Scope** – Summarizes the contributions of the project and outlines possible improvements and future research directions for enhancing the system.

1.9 Summary of Chapter

This chapter introduced the Dark Web Threat Monitoring Crawler and discussed the limitations of traditional cybersecurity monitoring tools in detecting threats originating from hidden web environments. It explained the need for automated crawling and analysis systems capable of extracting actionable intelligence from dark web sources.

The chapter outlined the problem definition, scope, objectives, challenges, assumptions, and societal relevance of the proposed system. By integrating Tor-enabled crawling, natural language processing, and machine learning-based classification, the project establishes a comprehensive framework for identifying and monitoring cyber threats emerging from the dark web. The next chapter reviews related research and existing approaches in dark web monitoring and threat intelligence.

Chapter 2

Literature Review

2.1 Introduction

The rapid expansion of dark web activities has necessitated the development of automated systems for monitoring and detecting cyber threats. The dark web, which operates over anonymity networks such as Tor [1], hosts marketplaces, forums, and communication channels that often facilitate malicious activities including phishing, credential leaks, and data trading [2, 3, 4]. This literature survey explores significant research works that focus on dark web crawling, threat data extraction, and machine learning-based analysis [5, 6]. The review aims to understand existing frameworks, their methodologies, and identify limitations that justify the design of the proposed *Dark Web Threat Monitoring Crawler*.

2.2 Survey Papers

Several noteworthy systems have been proposed in academic and industrial research to monitor and analyze dark web data. The most relevant papers are summarized below.

2.2.1 CRATOR: A Crawler for TOR—Turning Dark Web Pages into OSINT

CRATOR presents a framework for extracting content from Tor websites and converting it into structured Open Source Intelligence (OSINT) [2]. The crawler was designed to gather hidden web data for cybersecurity intelligence purposes. It uses a modular approach to parse and store Tor data. However, CRATOR struggles with dynamic content and unstructured HTML, leading to incomplete extraction. Frequent structural changes in dark web websites also require manual reconfiguration of crawling rules, affecting its adaptability.

2.2.2 DarkLens: A Conceptual Tool for Advancing OSINT through Dark Web

DarkLens proposes a conceptual model for enhancing OSINT collection using dark web data [7]. It emphasizes the importance of structured data extraction and intelligence enrichment. The study focuses primarily on design-level architecture rather than real-world deployment. Although the tool provides a strong conceptual foundation for OSINT data analysis, it lacks implementation results, evaluation metrics, and scalability validation. This limits its practical usability in large-scale monitoring.

2.2.3 BlackWidow: Monitoring the Dark Web for Cyber Security Information

BlackWidow is a continuous dark web monitoring system that collects data from forums and marketplaces [3]. It focuses on identifying emerging cyber threats, leaked credentials, and malware trade activity. The system achieves high-frequency crawling and incremental data updates to maintain relevance. Despite its continuous monitoring capability, the study highlights performance challenges related to large-scale data ingestion and limited source coverage. The system also lacks advanced NLP-based classification, relying primarily on keyword detection [5].

2.2.4 THREAT/crawl: A Trainable, Highly-Reusable Crawler for Underground Forums

THREAT/crawl introduces a trainable, modular crawler that can adapt to changing structures of underground forums [4]. The model allows retraining based on structural shifts, improving reusability and maintainability. While innovative, the crawler struggles with handling human-like interactions and authentication-based content. It also lacks an automated severity assessment mechanism and visualization components for threat prioritization.

2.2.5 Dark Web Threat Intelligence: Machine Learning-Based Approach for Cyber Threat Detection

This study introduces an ML-driven framework for detecting and classifying cyber threats from dark web textual data using Natural Language Processing (NLP) techniques [5]. It focuses on identifying malicious content such as phishing attempts, credential leaks,

and impersonation activities. The findings demonstrate that transformer-based architectures like BERT considerably enhance detection accuracy over traditional rule-based approaches. Nonetheless, the framework’s effectiveness is limited by its dependence on large labeled datasets and its lack of adaptability to the dynamic, real-time nature of dark web environments.

2.3 Comparison and Gaps Identified

Table 2.1 provides a comparative summary of the reviewed systems based on objectives, methodologies, and limitations.

Paper / System	Objective	Methodology	Limitations
CRATOR (2019) [2]	Convert dark web pages into OSINT data	Crawler-based extraction and structuring of Tor data	Poor handling of dynamic content, requires frequent manual updates
DarkLens (2020) [7]	Conceptual tool for dark web OSINT	Structured data extraction and conceptual analysis model	Not practically implemented or tested on real data
BlackWidow (2021) [3]	Continuous monitoring of dark web for cybersecurity information	Automated crawler collecting data from forums and marketplaces	Limited scalability and no ML-based threat classification
THREAT/crawl (2021) [4]	Automate collection from criminal underground forums	Modular and trainable crawler for structural adaptation	Requires retraining; lacks human-like interaction handling and visualization
Dark Web Threat Intelligence (2022) [5]	Apply ML and NLP to classify dark web threats	Transformer-based classification (BERT / DarkBERT)	Needs large labeled dataset; lacks integration with live crawling

Table 2.1: Comparison of Existing Dark Web Monitoring and Threat Detection Systems

Identified Gaps

From the comparison above, the following research gaps are evident:

- **Limited End-to-End Automation:** Most systems focus either on crawling or analysis, lacking a fully integrated threat monitoring pipeline.
- **Absence of Real-Time Threat Classification:** Few systems leverage NLP and ML in real time to detect and score potential threats.
- **Lack of Severity Assessment Models:** None of the reviewed studies provide a quantitative measure of threat severity or prioritization.
- **Inadequate Visualization Tools:** Reporting and visualization features are often overlooked, limiting usability for security analysts.
- **Poor Adaptability:** Frequent dark web structural changes require retraining or reconfiguration in existing crawlers.

Proposed System Advantage

The proposed *Dark Web Threat Monitoring Crawler* bridges these gaps by combining:

- A Tor-enabled focused crawler for secure, targeted data collection [1].
- NLP-driven preprocessing and ML-based classification using transformer models (DarkBERT / DistilBERT) [5].
- A hybrid severity scoring algorithm integrating keyword, confidence, and recency factors.
- A Streamlit-based visualization dashboard for real-time monitoring and reporting.

This integrated approach ensures automation, adaptability, and practical usability in cybersecurity environments.

2.4 Summary of Chapter

This chapter presented a comprehensive survey of existing literature in the domain of dark web crawling and threat monitoring. The comparative analysis shows that while earlier systems laid important foundations [2, 7, 3, 4, 5, 1], they fall short in terms of automation, scalability, and analytical sophistication. By incorporating focused Tor-based crawling, NLP-driven classification, and an interactive dashboard, the proposed system aims to provide a more robust, intelligent, and accessible solution for early detection of dark web threats.

Chapter 3

System Design

This chapter presents the design and architecture of the Dark Web Threat Monitoring Crawler. The system is designed as a modular pipeline that integrates Tor-based web crawling, data preprocessing, threat detection, machine learning classification, and visualization components. Each module performs a specific task in the overall workflow, enabling secure data acquisition, intelligent threat analysis, and interactive visualization of detected threats. The design emphasizes scalability, modularity, and efficient data processing so that each component can be independently improved without affecting the overall system.

3.1 Overall Architecture

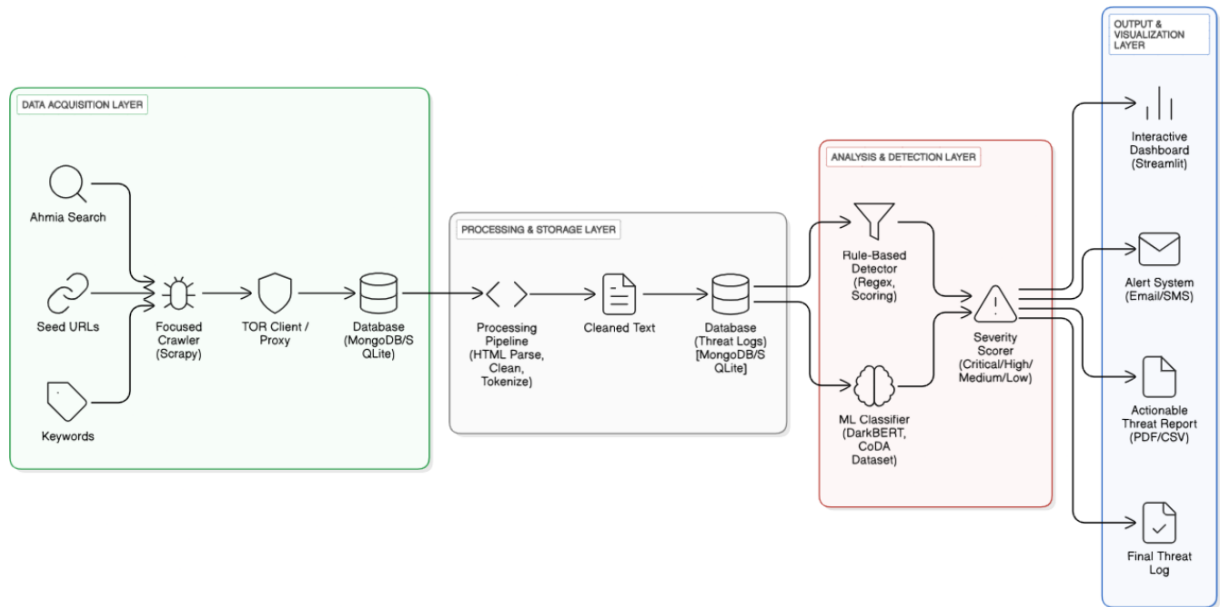


Figure 3.1: System Architecture of the Dark Web Threat Monitoring Crawler

The overall architecture of the Dark Web Threat Monitoring Crawler follows a layered design that transforms raw dark web data into actionable threat intelligence, as shown in Fig. 3.1. The system consists of four primary layers: the data acquisition layer, preprocessing layer, threat detection and classification layer, and visualization layer.

The data acquisition layer is responsible for crawling dark web sources through the Tor network. It retrieves .onion links using Ahmia search and collects web page content through a Tor-enabled crawler. The retrieved HTML pages are then passed to the preprocessing layer, where the raw content is cleaned and converted into structured textual data.

The threat detection and classification layer analyzes the processed text using both rule-based detection methods and machine learning models. Rule-based techniques identify explicit threat indicators such as credential leaks or phishing patterns, while transformer-based models provide contextual classification of potential threats. Finally, the visualization layer presents the detected threats through an interactive dashboard, enabling security analysts to monitor trends, filter results, and generate reports.

3.2 Component Design

The system is composed of several major components that work together to perform automated dark web monitoring and threat detection. Each component is designed to perform a specific task while interacting with other components through clearly defined interfaces.

3.2.1 Dark Web Crawler

The dark web crawler is responsible for collecting data from hidden web services. It operates through the Tor network using a SOCKS proxy to maintain anonymity. The crawler retrieves .onion URLs from Ahmia search and scans relevant pages using organization-specific keywords. The extracted HTML content and metadata are stored for further processing.

3.2.2 Preprocessing Module

The preprocessing module converts raw HTML content into structured textual data suitable for analysis. It removes HTML tags, scripts, and other irrelevant elements before applying text normalization techniques. The module performs tokenization, stop-word removal, and text cleaning to produce meaningful input for the threat detection system.

3.2.3 Threat Detection and Classification Module

This module identifies potential cybersecurity threats within the processed text. Rule-based detection mechanisms identify explicit indicators such as credential leaks, cryptocurrency wallet addresses, and suspicious login pages. Additionally, a machine learning classifier based on transformer models such as DarkBERT analyzes the contextual meaning of the text and categorizes it into threat types such as credential leak, phishing attempt, impersonation, or benign content.

3.2.4 Visualization Dashboard

The visualization component presents the analyzed results through an interactive dashboard developed using Streamlit. The dashboard displays threat summaries, severity distribution charts, and keyword frequency statistics. Users can filter threats by organization, severity level, or time period and export reports for further analysis.

3.3 Module Division

The system is divided into four primary modules:

- **Dark Web Crawling Module** Responsible for accessing .onion sites through the Tor network and collecting relevant web page content.
- **Data Preprocessing Module** Processes raw HTML content and converts it into normalized textual data suitable for analysis.
- **Threat Detection and Classification Module** Analyzes textual content using rule-based detection techniques and machine learning models to identify potential cybersecurity threats.

- **Visualization and Alert Module** Displays detected threats through an interactive dashboard and generates alerts or reports for security analysts.

3.4 Tools and Technologies

The implementation of the Dark Web Threat Monitoring Crawler uses several software and hardware tools.

Software Requirements

- Programming Language: Python
- Web Crawling Libraries: Requests, BeautifulSoup
- Tor Integration: Stem library and Tor client
- Natural Language Processing: spaCy / NLTK
- Machine Learning Framework: Transformers (DarkBERT)
- Dashboard Development: Streamlit
- Database: SQLite / MongoDB
- Operating System: Linux / Windows

Hardware Requirements

- Minimum 8 GB RAM
- Multi-core processor
- Stable internet connectivity
- Secure storage for crawled data and models

3.5 Work breakdown and Responsibilities

The project work is divided among the team members to ensure efficient development and integration of system components.

- **Diya Vypana** – Responsible for implementing the Tor-enabled crawler, configuring the crawling environment, and integrating data collection modules.
- **Mathew Dominic** – Responsible for database design, data extraction pipelines, and implementing rule-based threat detection mechanisms.
- **Levana P Saju** – Responsible for developing the natural language processing pipeline and implementing machine learning models for threat classification.
- **Jithin Shabu** – Responsible for developing the visualization dashboard, implementing threat analytics, and generating reports and alerts.

3.6 Key Deliverables

The key deliverables of the project include:

- A Tor-enabled dark web crawler capable of retrieving .onion web pages.
- A preprocessing pipeline for cleaning and extracting textual information from crawled data.
- A hybrid threat detection system combining rule-based methods and machine learning classification.
- A structured database for storing detected threats and metadata.
- An interactive dashboard for visualizing threats and generating reports.
- Documentation and evaluation of system performance.

3.7 Project Timeline

The project development was carried out in several stages:

- **Phase 1 – Research and System Design** Literature review, architecture design, and selection of tools and technologies.
- **Phase 2 – Core Development** Implementation of the Tor-based crawler and preprocessing pipeline.

- **Phase 3 – Threat Detection Implementation** Development of rule-based detection mechanisms and integration of machine learning classification models.
- **Phase 4 – Visualization and Integration** Development of the Streamlit dashboard and integration of all modules.
- **Phase 5 – Testing and Documentation** System testing, evaluation, and preparation of the final project report.

3.8 Summary of chapter

This chapter presented the design and architecture of the Dark Web Threat Monitoring Crawler. The overall system architecture, major components, module division, and development tools were discussed in detail. The chapter also described the work distribution among team members and the timeline followed during project development. The next chapter focuses on the implementation details and results obtained from the developed system.

Chapter 4

Implementation

This chapter describes the implementation of the Dark Web Threat Monitoring Crawler. It explains how the different modules of the system were developed and integrated to form a complete pipeline for collecting, analyzing, and visualizing potential cybersecurity threats from the dark web. The implementation focuses on secure crawling through the Tor network, preprocessing of collected data, threat detection using both rule-based and machine learning techniques, and visualization of the results through an interactive dashboard.

4.1 Crawler Implementation

The crawling module is responsible for collecting data from dark web sources. The crawler operates through the Tor network using a SOCKS proxy to ensure anonymity while accessing .onion websites. Onion links are obtained using Ahmia search, which indexes publicly accessible dark web pages and provides a starting point for discovering hidden services.

The crawler sends requests through the Tor network and retrieves the HTML content of the target web pages. Python libraries such as `requests` and `BeautifulSoup` are used for web requests and HTML parsing. To avoid excessive network load and detection by hidden services, the crawler implements rate limiting and depth restrictions. These limitations ensure that the crawler behaves responsibly and avoids generating large volumes of requests that could overload servers.

The collected HTML pages, along with metadata such as URL, crawl timestamp, and page source, are stored for further processing. This information allows the system to track the origin of the detected threats and perform additional analysis when necessary.

4.2 Tor Network Integration

The crawler communicates with dark web services through the Tor network in order to access .onion domains securely and anonymously. A Tor client runs locally and exposes a SOCKS5 proxy interface. All HTTP requests generated by the crawler are routed through this proxy using the `requests` library configured with the SOCKS protocol.

The Stem library is used to manage Tor circuits and maintain reliable connectivity. If the crawler encounters connection errors or slow responses, new Tor circuits can be requested to improve routing stability. This approach helps the crawler maintain anonymity while ensuring successful communication with hidden services.

Using the Tor network also ensures that domain resolution for .onion addresses occurs inside the Tor environment rather than through standard DNS systems. This configuration is necessary because .onion domains are not accessible through conventional internet infrastructure.

4.3 Preprocessing Module

After collecting HTML pages from dark web sources, the preprocessing module converts raw web content into structured textual data suitable for analysis. Dark web pages often contain noisy HTML structures, embedded scripts, navigation elements, and formatting tags that do not contribute to meaningful analysis.

The preprocessing module uses the BeautifulSoup library to parse the HTML content and remove unnecessary elements such as script tags, style tags, and hidden content. The remaining visible text is extracted and cleaned to obtain readable information.

The extracted text is then normalized through several Natural Language Processing techniques. These include tokenization, removal of stop words, and text normalization. These preprocessing steps reduce noise and ensure that the textual data is suitable for further analysis by the threat detection module.

4.4 Threat Detection and Classification

The threat detection module analyzes the processed textual data to identify potential cybersecurity threats. A hybrid detection strategy is used, combining rule-based techniques

with machine learning classification.

The rule-based detection system relies on pattern matching and regular expressions to detect explicit indicators of malicious activity. These indicators may include leaked credentials, suspicious login pages, references to stolen databases, cryptocurrency wallet addresses, or mentions of sensitive organizational information. Because these patterns are deterministic, they allow the system to quickly identify high-confidence threat signals.

In addition to rule-based detection, the system incorporates a machine learning classifier based on transformer models such as DarkBERT. Transformer models provide contextual understanding of textual data and can identify threats that are not easily detectable through simple pattern matching. The classifier analyzes the context of sentences and categorizes content into classes such as credential leaks, phishing attempts, impersonation attacks, marketplace listings, or benign mentions.

The results from the rule-based system and the machine learning classifier are combined to determine the final threat classification. Each detected threat is assigned a severity level such as low, medium, or high based on the confidence of the detection and the type of indicators present in the content. This hybrid approach improves detection accuracy while reducing false positives.

4.5 Database Storage

All collected and processed data are stored in a structured database to support further analysis and visualization. The database stores information such as crawled URLs, extracted text, timestamps, detection results, and threat severity levels.

Storing this information allows the system to maintain historical records of detected threats and enables efficient querying by the visualization module. The database also allows the system to track trends in dark web activity over time and helps analysts investigate previously detected incidents.

The stored records serve as the primary data source for the dashboard interface and reporting system. By maintaining structured storage of crawl results and threat data, the system ensures that analysis can be reproduced and verified without repeatedly crawling the same pages.

4.6 Visualization Dashboard

The results generated by the threat detection system are presented through an interactive dashboard developed using Streamlit. The dashboard provides a user-friendly interface for monitoring detected threats and analyzing cybersecurity trends.

The dashboard displays summary statistics such as the number of detected threats, threat category distribution, and severity levels. Graphical visualizations including charts and tables help users understand patterns in the collected data. These visual elements allow analysts to quickly identify spikes in malicious activity or emerging trends related to specific keywords or organizations.

Users can filter threats based on parameters such as organization name, time period, severity level, or keyword relevance. This filtering capability enables targeted investigation of suspicious activity. The dashboard also supports exporting reports in formats such as CSV or PDF, allowing analysts to review and document detected threats for further investigation.

4.7 Summary of Chapter

This chapter described the implementation of the Dark Web Threat Monitoring Crawler and explained how its major modules were developed and integrated. The Tor-enabled crawler was implemented to collect relevant .onion pages, while the preprocessing pipeline transformed raw HTML content into structured textual data suitable for analysis. A hybrid threat detection approach combining rule-based pattern matching and transformer-based machine learning classification was applied to identify potential cyber threats. The results were then presented through an interactive Streamlit dashboard that allows users to monitor threat trends and generate reports. The next chapter presents the experimental results and discusses the performance and effectiveness of the proposed system.

Chapter 5

Results and Discussions

This chapter presents the results obtained from the implementation of the Dark Web Threat Monitoring Crawler and discusses the overall performance of the system. The developed framework integrates Tor-based crawling, data preprocessing, hybrid threat detection, and visualization components to monitor potential cybersecurity threats from dark web sources. The results demonstrate how the crawler collects data from hidden services, how threats are detected using both rule-based and machine learning techniques, and how the extracted information is presented through the visualization dashboard.

The primary objective of this evaluation is to verify the effectiveness of the proposed system in identifying potential threats from dark web content and providing meaningful insights for cybersecurity monitoring.

5.1 Threat Detection Results

The implemented crawler successfully retrieved data from publicly accessible `.onion` websites through the Tor network. Using Ahmia search as a source of seed URLs, the crawler was able to discover and access relevant hidden services containing potentially suspicious or security-related content. The retrieved pages were stored along with metadata such as the source URL and crawl timestamp.

After crawling, the preprocessing module processed the raw HTML content and converted it into structured textual data. HTML tags, scripts, stylesheets, and other irrelevant elements were removed using the BeautifulSoup parsing library. This step ensured that only meaningful textual information remained for analysis.

The hybrid threat detection system was then applied to the processed text. The rule-based detection module successfully identified explicit threat indicators such as credential patterns, suspicious login references, and potential data leak indicators. These rules

enabled the system to quickly detect clear signs of malicious activity.

Figure 5.1 shows an example of a detected threat along with the text snippet that triggered the rule-based detection mechanism. The system highlights suspicious indicators and displays the evidence snippet that caused the detection.

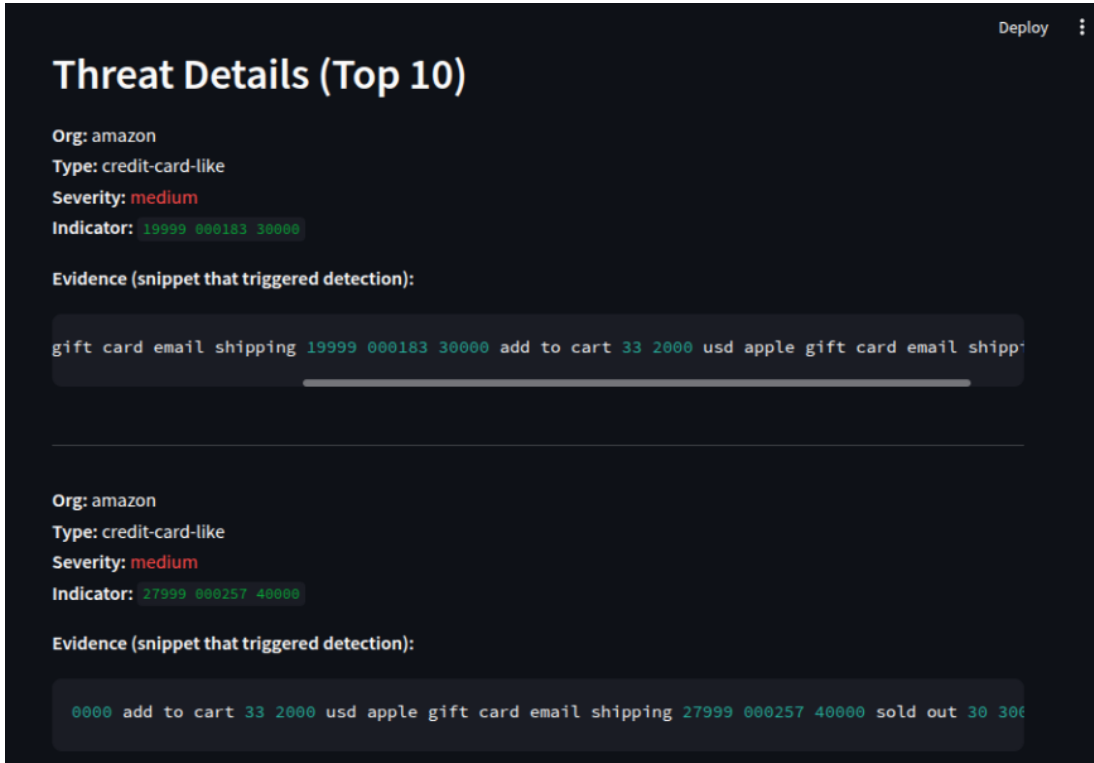


Figure 5.1: Example threat evidence snippet detected by the rule-based detection module

The detected threats are stored in the system database along with metadata such as the source URL, organization identifier, indicator type, and severity level. This structured storage enables further analysis and historical tracking of detected threats.

Figure 5.2 shows the stored crawl records and detected threats within the database. The table displays information including crawled URLs, HTTP response status, detection indicators, and severity classification.

Deploy

Recent Crawled Pages

Download

Search

Columns

	id	org_id	url	status_code	fetchd_
0	159	4	http://deepweb4wt3m4dhutpxpe7d7wxdfdf4hhag4sizgon6th5lcefloid.onion/	200	2025-12-
1	158	4	http://amnesia7u5odx5xbwtpnqk3edybgud5bmiagu75bnqx2crntw5kry7ad.onion/se	200	2025-12-
2	157	4	http://amnesia7u5odx5xbwtpnqk3edybgud5bmiagu75bnqx2crntw5kry7ad.onion/se	200	2025-12-
3	156	4	http://2fd6cemt4gmccflhm6imvdfvli3nf7zn6rfrwpsy7uhxrgbypvwf5fad.onion/search	200	2025-12-
4	155	4	http://b26lqoc5u35asqnlka254bsn2ifq2xzw23azfe3oovxf35enwestjyd.onion	200	2025-12-
5	34	4	http://amazoniallve47iexufcqqccj6nufmieytnvcf6ykdcet3hix2nj2yd.onion	200	2025-12-
6	33	4	http://torlbmqwtudkorme6prgfpmsnile7ug2zm4u3ejpcncxuhpu4k2j4kyd.onion/inde	200	2025-12-
7	32	4	http://torlbmqwtudkorme6prgfpmsnile7ug2zm4u3ejpcncxuhpu4k2j4kyd.onion/inde	200	2025-12-
8	31	4	http://torlbmqwtudkorme6prgfpmsnile7ug2zm4u3ejpcncxuhpu4k2j4kyd.onion	200	2025-12-
9	30	4	http://juhanurmihxlp77nkq76byazcldy2hlmovfu2epvl5ankdibsot4csyd.onion/	200	2025-12-

Detected Threats

	id	org_id	org_name	crawled_page_id	indicator_type	indicator	severity	evidence
0	320	4	amazon	156	credit-card-like	19999 000183 30000	medium	000092 add to cart 1000

Figure 5.2: Database records showing crawled pages and detected threats

5.2 Visualization and Dashboard Analysis

To enable effective monitoring of the detected threats, the results were visualized using an interactive dashboard developed with the Streamlit framework. The dashboard provides a centralized interface where users can explore and analyze threat intelligence extracted from dark web sources.

The dashboard displays several key metrics including the number of crawled pages, detected threats, and the distribution of threat categories. Graphical visualizations such as charts and tables allow users to quickly understand patterns in the collected data.

Figure 5.3 illustrates the distribution of threat categories detected by the machine learning classifier. The chart shows the proportion of different types of detected threats such as scams, marketplaces, forums, and credential leaks.

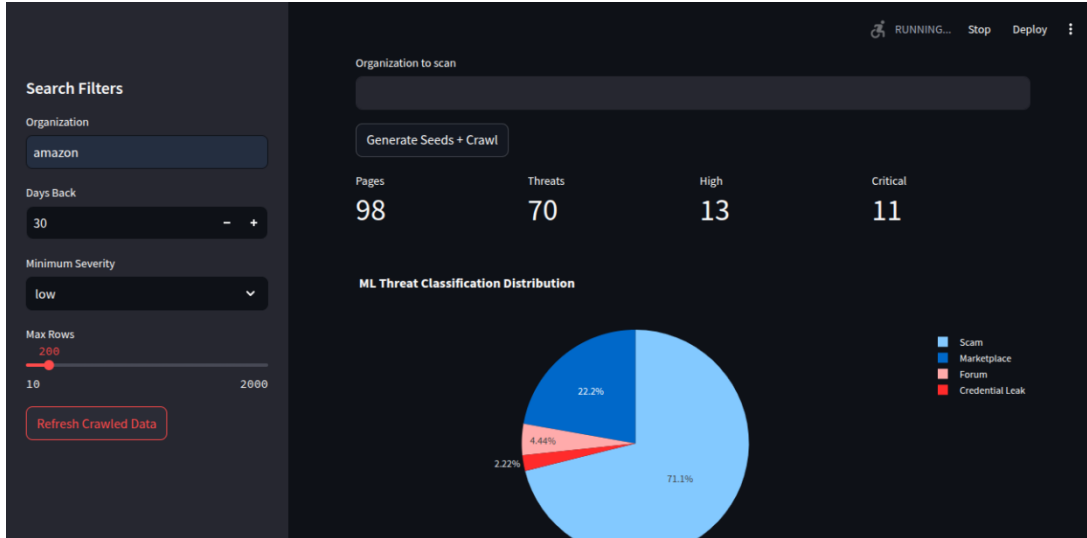


Figure 5.3: Threat classification distribution displayed in the Streamlit dashboard

Users can filter threats based on parameters such as organization name, time period, severity level, or keyword relevance. This filtering capability enables targeted investigation of suspicious activity. Additionally, the dashboard provides the ability to export results as structured reports in formats such as CSV or PDF. These reports can be used for documentation, incident analysis, or sharing threat intelligence with security teams.

5.3 System Performance Discussion

The overall performance of the system was evaluated based on its ability to reliably crawl dark web sources, preprocess extracted data, and accurately identify potential threats.

The Tor-enabled crawler demonstrated stable performance when accessing hidden services. Despite the inherent latency and instability associated with the Tor network, the crawler was able to successfully retrieve relevant pages while maintaining anonymity. Rate limiting and controlled crawling depth helped ensure responsible operation and prevented excessive network load.

The preprocessing module effectively improved the quality of the textual data extracted from crawled pages. By removing irrelevant HTML elements and normalizing the text, the module ensured that the threat detection system received clean and structured input.

The hybrid detection approach proved to be particularly effective. Rule-based detection allowed rapid identification of explicit threat indicators, while the machine learning

classifier provided deeper contextual understanding of the text. This combination improved detection reliability and reduced the likelihood of missing subtle threats.

The Streamlit-based dashboard also performed effectively as a visualization interface. It allowed analysts to easily interpret detection results and monitor dark web activity through an intuitive interface.

5.4 Summary of Results

The experimental results demonstrate that the proposed Dark Web Threat Monitoring Crawler successfully performs automated monitoring of dark web sources. The Tor-enabled crawler effectively retrieves content from `.onion` websites, while the preprocessing module transforms raw HTML pages into structured textual data suitable for analysis.

The hybrid threat detection mechanism combining rule-based pattern matching and transformer-based machine learning classification enables accurate identification of potential cybersecurity threats. Detected threats are stored with relevant metadata and presented through an interactive dashboard that supports filtering, visualization, and reporting.

These results confirm that the proposed system provides a practical and effective approach for monitoring dark web activity and identifying emerging cyber threats.

Chapter 6

Conclusions & Future Scope

6.1 Conclusion

This project presented the design and implementation of a Dark Web Threat Monitoring Crawler aimed at identifying potential cybersecurity threats originating from hidden web networks. The system integrates Tor-enabled crawling, data preprocessing, rule-based threat detection, and machine learning classification to automatically analyze content collected from .onion websites.

By extracting and processing textual information from dark web pages, the proposed system is capable of detecting indicators of threats such as credential leaks, phishing activities, and impersonation attempts. The hybrid detection approach combining rule-based techniques and transformer-based machine learning models improves the ability to identify both explicit and contextual threat patterns.

In addition, the integration of an interactive visualization dashboard enhances the usability of the system by allowing security analysts to monitor threat trends, filter detected threats, and generate reports for further investigation. The results demonstrate that automated crawling and intelligent data analysis can effectively support cyber threat intelligence and provide early warning signals for organizations facing potential risks from dark web sources.

Overall, the project validates the feasibility of combining web crawling, natural language processing, and machine learning techniques to develop an automated system for monitoring and analyzing dark web threats.

6.2 Future Scope

Although the proposed system provides a functional framework for dark web threat monitoring, several enhancements can be explored in future work. The crawling module can

be extended with parallel or distributed crawling techniques to improve scalability and coverage when scanning a larger number of dark web pages.

More advanced machine learning models and larger training datasets could be incorporated to further improve the accuracy of threat classification and reduce false positives. The system could also be enhanced with real-time alerting mechanisms that automatically notify organizations when high-severity threats are detected.

In addition, integrating graph-based analysis techniques and expanding support for dynamically changing dark web structures would further strengthen the system's capability as a scalable cyber threat intelligence platform. These improvements would enable the system to provide more comprehensive monitoring and faster detection of emerging threats in the dark web ecosystem.

References

- [1] T. T. Project, “Anonymity online: Understanding the tor network,” <https://www.torproject.org>, 2023.
- [2] D. De Pascale, G. Cascavilla, D. A. Tamburri, and W. J. Van Den Heuvel, “Crator: A crawler for tor—turning dark web pages into osint,” in *Proceedings of the IEEE International Conference on Big Data*, 2024.
- [3] M. Schäfer, D. A. Tamburri, and W. J. Van Den Heuvel, “Blackwidow: Monitoring the dark web for cyber security information,” *IEEE Transactions on Dependable and Secure Computing*, 2021.
- [4] M. Campobasso and L. Allodi, “Threat/crawl: Automated crawling of criminal underground forums,” in *Proceedings of the IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*, 2021.
- [5] K. K. Khaing and T. Thein, “Dark web threat intelligence: Machine learning-based approach for cyber threat detection,” *International Journal of Computer Applications*, 2022.
- [6] Y. Jin, E. Jang, Y. Lee, S. Shin, and J.-W. Chung, “Shedding new light on the language of the dark web,” arXiv preprint arXiv:2204.06106, submitted on 14 Apr 2022, last revised 4 May 2022 (v2).
- [7] A. Amin, “Features and implementation of darklens: A conceptual tool for advancing osint through dark web,” Department of Information Security, NUST MCS, Islamabad, Pakistan, Tech. Rep., 2020.
- [8] S. Mohammadi, A. A. Feizollah, and N. B. Anuar, “A survey of machine learning techniques for cybersecurity,” *IEEE Communications Surveys & Tutorials*, vol. 21, no. 4, pp. 3631–3671, 2019.

- [9] P. P. Shakarian, A. Shakarian, and A. Ruef, *Introduction to Cyber-Warfare: A Multi-disciplinary Approach*. Elsevier, 2013.

Appendix A: Presentation

Dark Web Threat Monitoring Crawler

Guide:
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Associate Professor
Dept. of CSE

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Levana P Saju - U2203133
Mathew Dominic- U2203138

Purpose & Need

- The dark web is used for selling stolen data, phishing, and impersonation.
- Traditional security tools lack visibility into hidden Tor (.onion) networks.
- Need for an automated crawler to detect threats related to organizational data and identity.
- Enables early detection of data leaks, phishing, and impersonation attempts.
- Bridges the gap between conventional monitoring tools and the anonymous dark web.

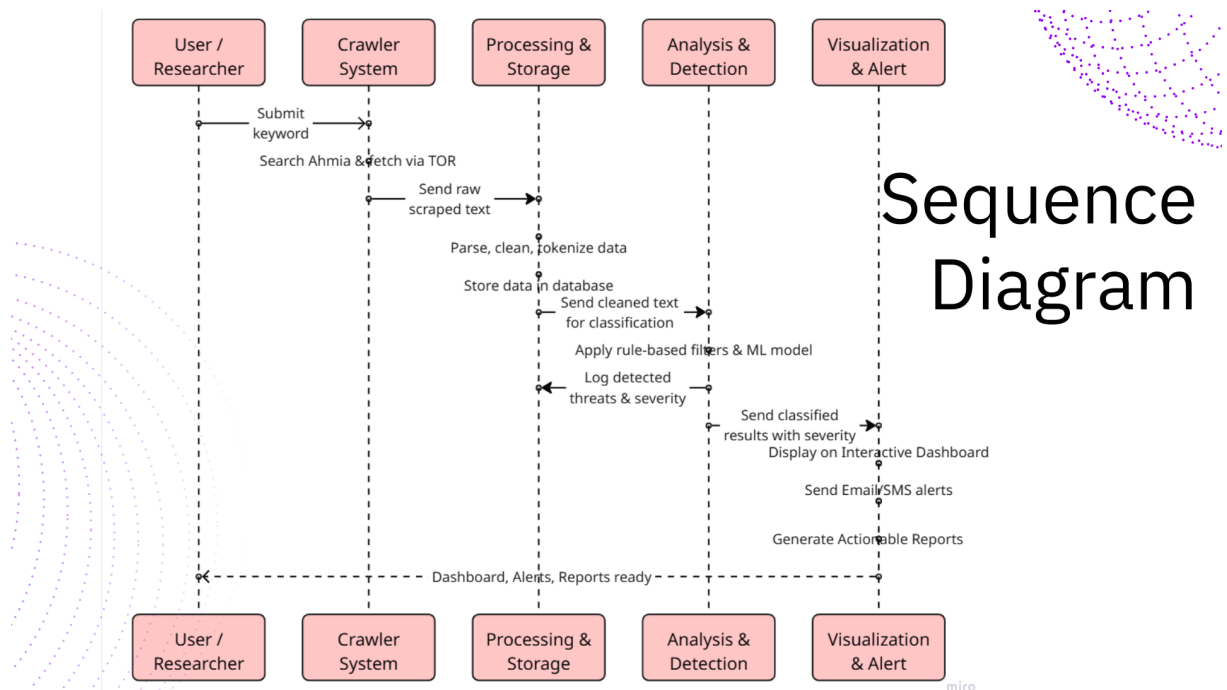
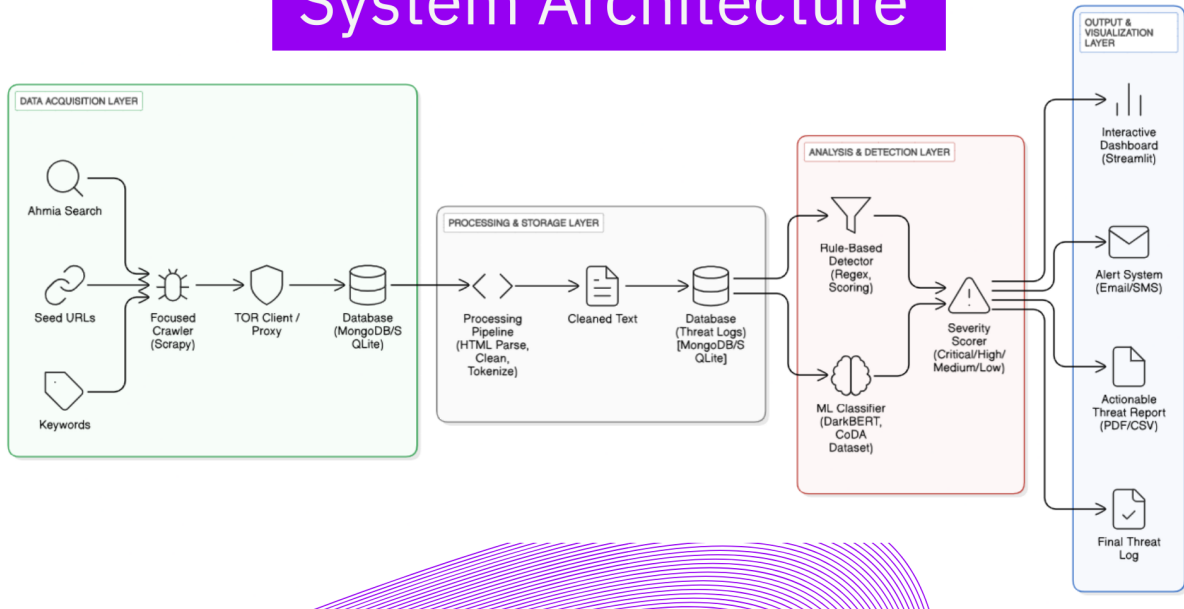
Project Objective

- To develop a Tor-enabled web crawler capable of scanning .onion sites securely and efficiently.
- To detect dark web threats such as phishing portals, impersonation attempts, and credential/data leaks.
- To analyze & classify crawled data using Natural Language Processing (NLP) and Machine Learning (ML) techniques to assess the severity of threats.
- To visualize the collected information through an interactive dashboard and generate detailed, exportable threat reports.
- To provide actionable, proactive insights to enhance the organization's cybersecurity preparedness.

Proposed Method

1. Tor-Enabled Crawling: Securely access and crawl .onion sites using Tor based on organization keywords.
2. Data Extraction & Preprocessing: Clean, tokenize, and store dark web text data for analysis.
3. Threat Detection and Classification: Use rule-based filters to identify phishing kits, impersonations, and data leaks. Apply DarkBERT to classify threats by type and severity.
4. Visualization & Reporting: Display results on an interactive Streamlit dashboard with reports.

System Architecture



Assumptions

- The Tor network remains accessible for crawling .onion sites.
- Target organizations provide valid identifiers (name, domains, brand terms).
- Publicly available .onion content (via Ahmia) is sufficient to detect major threats.
- Ethical and legal boundaries are respected during crawling.

Risk & Challenges

- Dark web sites frequently shut down or change URLs, which may affect crawler coverage.
- There is a risk of encountering malicious scripts or honeypots while crawling.
- The collected data may contain noise, leading to false positives in threat detection.
- Legal and ethical concerns may arise if the crawler unintentionally crosses restricted boundaries.

Modules



Dark Web Crawler



**Preprocessing and
Content filtering**

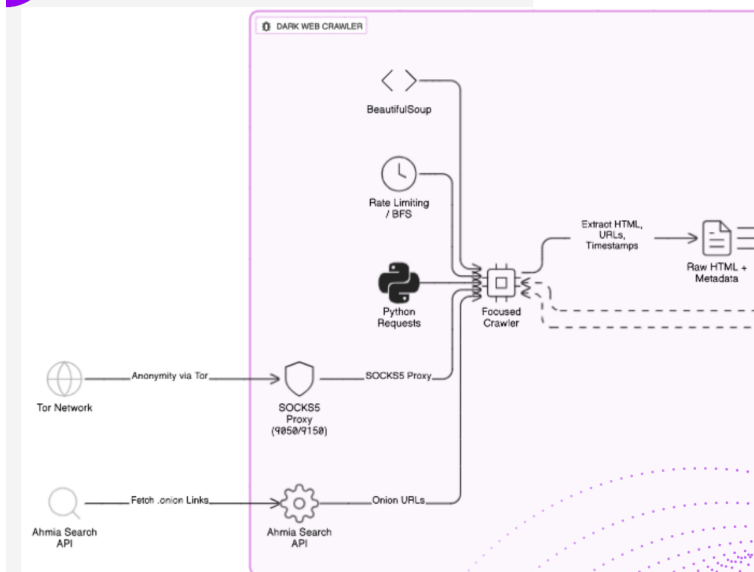


**Threat Detection
and Classification**



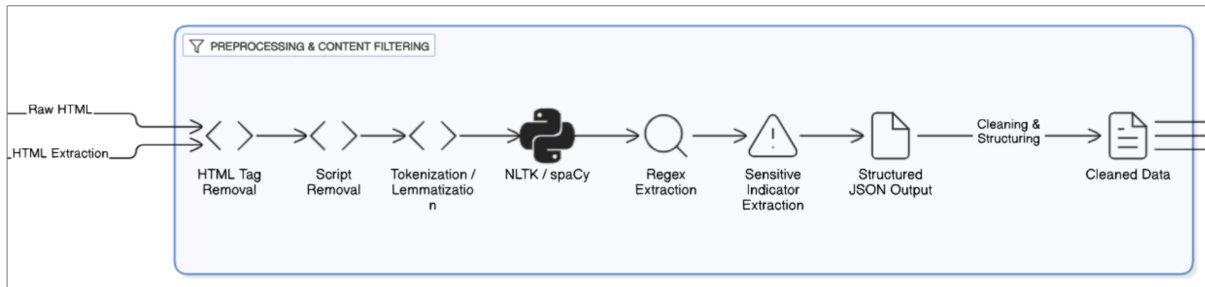
**Visualization and
Alert Dashboard**

Dark web Crawler



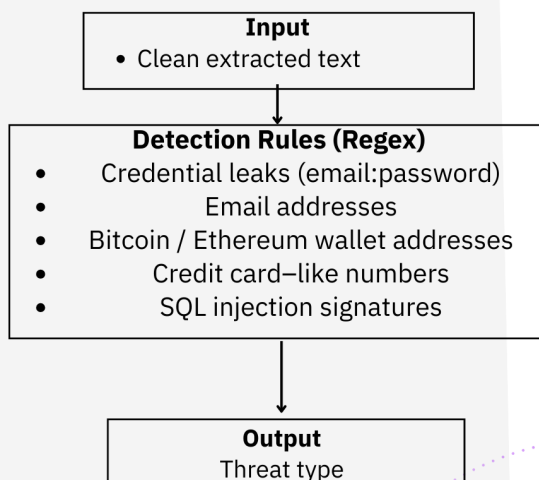
- Crawls dark web sources for domain, brand, or product data using Tor.
- Controls Tor circuits via Stem for reliable and anonymous access.
- Discovers relevant .onion links through the Ahmia Search.
- Extracts and stores HTML, text, and metadata using Python scrapers.

Preprocessing and Content filtering



- Cleans raw crawled HTML by removing tags, scripts, and irrelevant content.
- Normalizes text through tokenization, lemmatization, and stopword removal.
- Uses regex and NLP to extract sensitive indicators like credentials or phishing terms.
- Outputs structured, cleaned data for further threat detection and analysis.

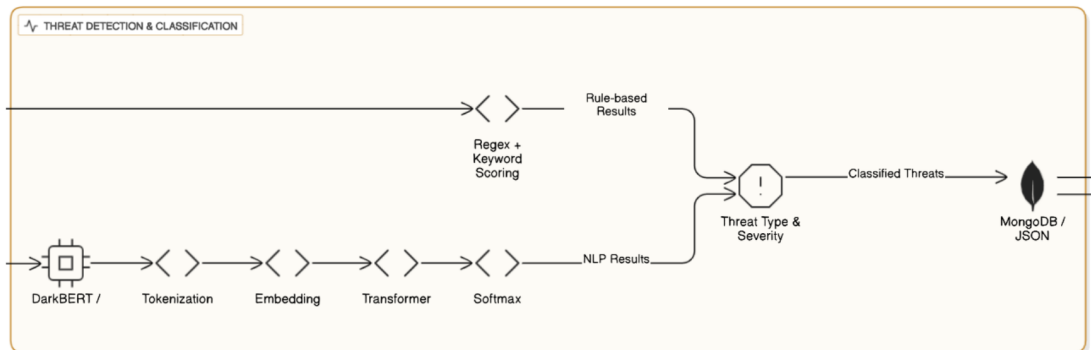
Rule-Based Threat Detection



Files:

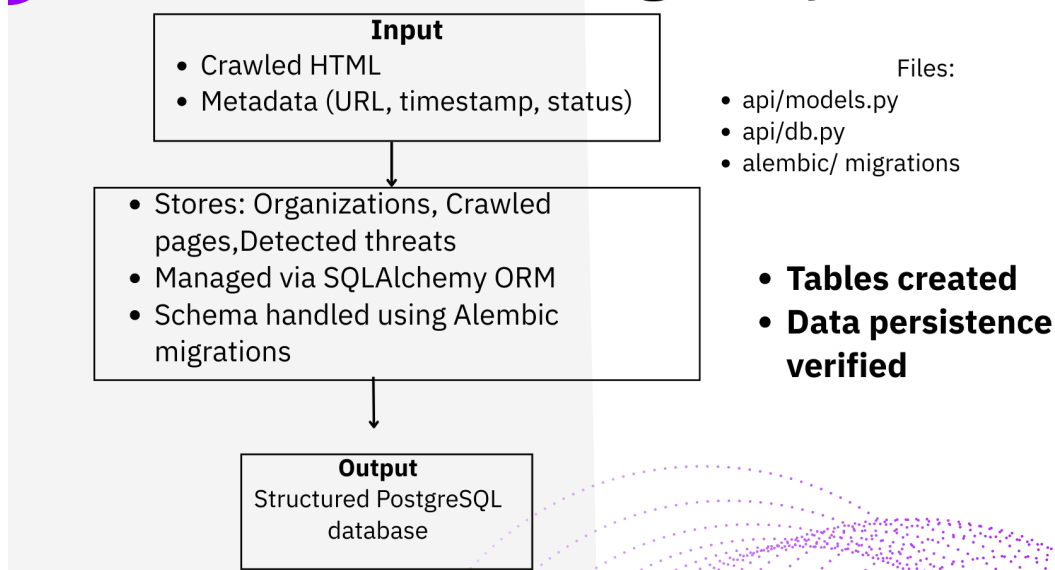
- services/preprocessor/detectors.py

Threat Detection & Classification



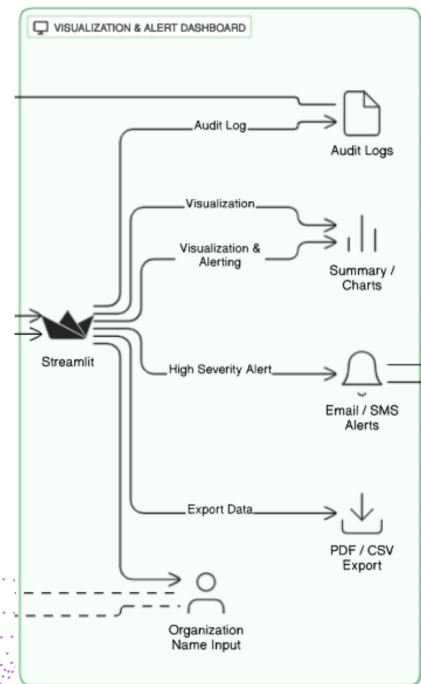
- Analyzes cleaned data using NLP-based models.
- Detects threat types like leaks or phishing.
- Assigns severity scores (Low, Medium, High).
- Stores classified results with threat details.

Database & Storage Layer

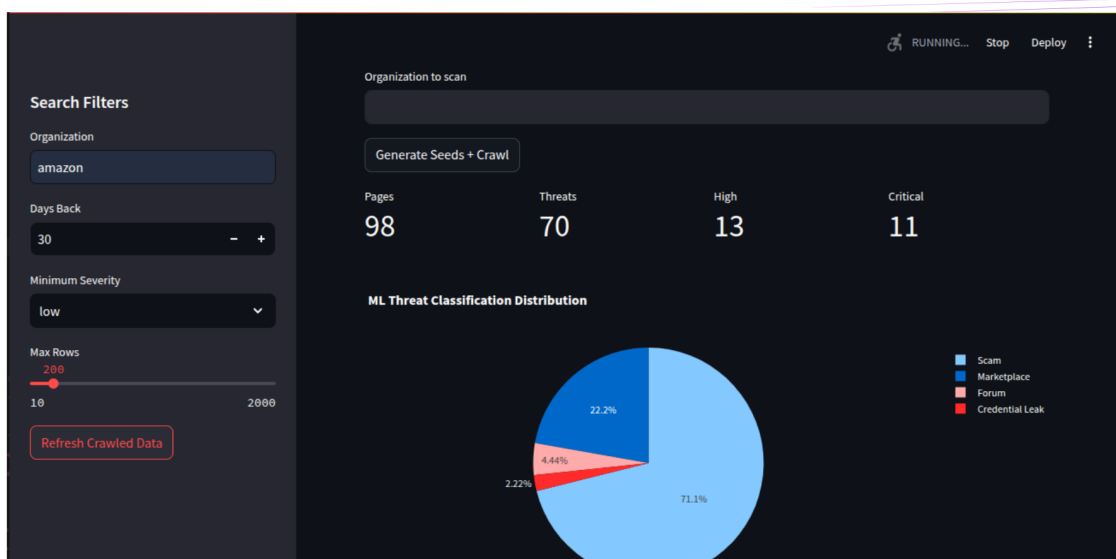


Visualization and Alert Dashboard

- Accepts organization inputs to customize monitoring and visual outputs.
- Displays analyzed threats and trends using Streamlit-based charts and summaries.
- Generates a threat summary report using Ollama model
- Generates audit logs and exports reports in PDF/CSV format.



UI DESIGN



UI DESIGN

Deploy

Recent Crawled Pages

	id	org_id	url	status_code	fetches
0	159	4	http://deepweb4wt3m4dhutpxpe7d7wxdftfd4fhag4sizon8th5lcefold.onion/	200	2025-12-
1	158	4	http://amnesia7u5odxsxbwtpnqk3edybgud5bmiagu75bnqx2cmtw5kry7ad.onion/se	200	2025-12-
2	157	4	http://amnesia7u5odxsxbwtpnqk3edybgud5bmiagu75bnqx2cmtw5kry7ad.onion/se	200	2025-12-
3	156	4	http://2fd6cent4gmccflhm6imvdlvli3nf7zn6frfwps7uhurgbyppvw5fad.onion/search	200	2025-12-
4	155	4	http://b26lqoc5u35asqnlka254bsn2lfq2xzw23azfe3oovxf35enwestjyd.onion	200	2025-12-
5	34	4	http://amazoniallve47ieufcqqccj6nufmiejvncf6ykdct3hj2nj2yd.onion	200	2025-12-
6	33	4	http://torlbmqwtudkorme6prgfpmsnile7ug2zm4u3ejpcncxuhpu4k2j4kyd.onion/ind	200	2025-12-
7	32	4	http://torlbmqwtudkorme6prgfpmsnile7ug2zm4u3ejpcncxuhpu4k2j4kyd.onion/ind	200	2025-12-
8	31	4	http://torlbmqwtudkorme6prgfpmsnile7ug2zm4u3ejpcncxuhpu4k2j4kyd.onion	200	2025-12-
9	30	4	http://juhanurmihxlp77nkq76byazcldy2hlmovfu2epvl5ankdibso4csyd.onion/	200	2025-12-

Detected Threats

	id	org_id	org_name	crawled_page_id	indicator_type	indicator	severity	evidence
0	320	4	amazon	156	credit-card-like	19999 000183 30000	medium	000092 add to cart 1000

UI DESIGN

Deploy

Threat Details (Top 10)

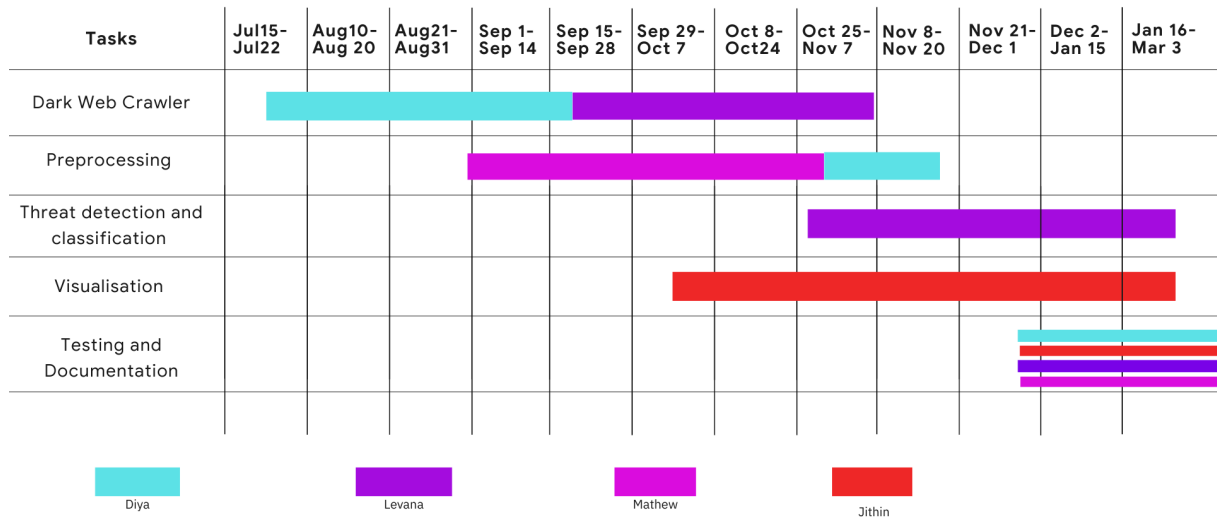
Org: amazon
Type: credit-card-like
Severity: medium
Indicator: 19999 000183 30000

Evidence (snippet that triggered detection):
gift card email shipping 19999 000183 30000 add to cart 33 2000 usd apple gift card email shipp

Org: amazon
Type: credit-card-like
Severity: medium
Indicator: 27999 000257 40000

Evidence (snippet that triggered detection):
0000 add to cart 33 2000 usd apple gift card email shipping 27999 000257 40000 sold out 30 300

Work Division - Gantt Chart



Resources Needed

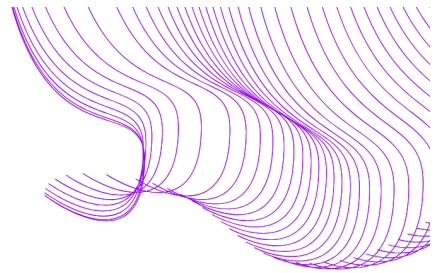
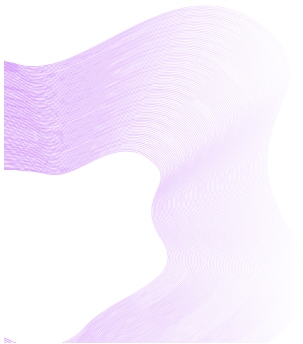
- Software Resources
 - Operating System: Linux (preferred) / Windows / macOS
 - Programming Language: Python
 - Tor Client: Tor daemon to access dark web (.onion) sites
 - Web Scraping: requests, BeautifulSoup
 - NLP Tools: spaCy or NLTK
 - Database: PostgreSQL
 - Dashboard: Streamlit
 - Ollama Local LLM inference
- Hardware Resources
 - Mid-range server/laptop with ≥ 8 GB RAM & stable internet.
 - Secure storage for crawled data.

Conclusion

The Dark Web Threat Monitoring Crawler enhances cybersecurity by proactively detecting hidden threats on the dark web. By leveraging Tor-based crawling, rule-based filtering, and machine learning, it enables early identification of phishing, impersonation, and data leaks. The system provides actionable intelligence through interactive dashboards and reports, empowering cybersecurity teams to respond quickly and strategically. Overall, it strengthens organizational resilience and contributes to a safer digital ecosystem.

References

1. D. De Pascale et al., CRATOR: A Crawler for TOR — Turning Dark Web Pages into OSINT.
2. A. Amin, DarkLens: A Conceptual Tool for Advancing OSINT through Dark Web.
3. M. Schäfer et al., BlackWidow: Monitoring the Dark Web for Cyber Security Information.
4. M. Campobasso and L. Allodi, THREAT/crawl: Automated Crawling of Criminal Underground Forums
5. K. K. Khaing and T. Thein. "Dark Web Threat Intelligence: Machine Learning-Based Approach for Cyber Threat Detection." Explores NLP and ML models for classifying and analyzing dark web threat data.



Thank **you**

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions for complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design or develop systems, components or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and the World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion, and adhere to national and international laws.

PO8: Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse or multidisciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, including the ability to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for (i) independent and life-long learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

COURSE OUTCOMES (CO):

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology or research gaps and propose innovative or creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

Appendix C: CO-PO-PSO Mapping

CO - PO - PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

Table 6.1: Justifications for CO-PO Mapping

Mapping	Level	Justification
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.

Mapping	Level	Justification
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.

Mapping	Level	Justification
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.

Mapping	Level	Justification
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.